

u5-optical_fiber

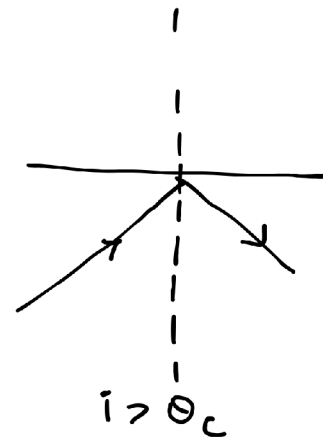
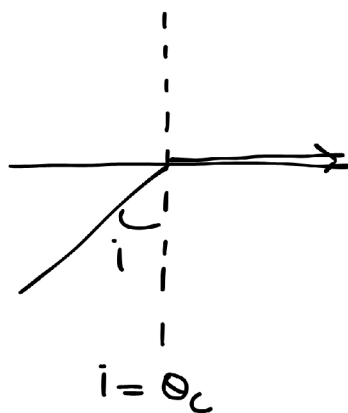
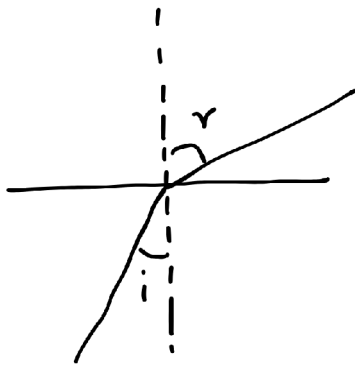
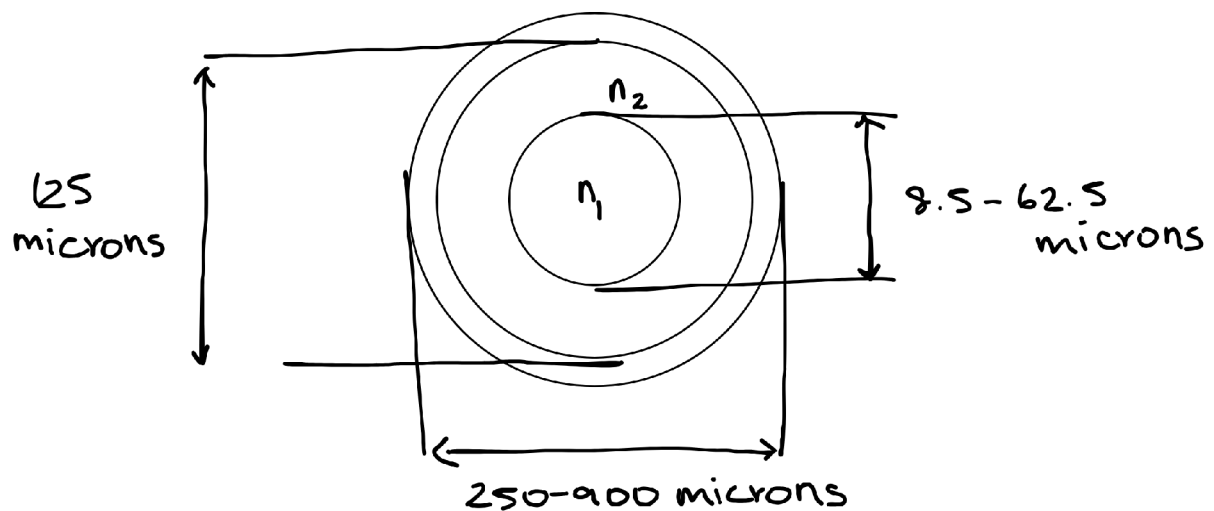
n01-advantages

- It is a dielectric material which guides light through it in optical frequency range.

Advantages over metal wires

1. cheaper
2. Not hazardous
3. Immune to EMI & RFI
EMI - Electromagnetic Interference
RFI - Radio Frequency Interference
4. No cross talk
5. Lightweight, strong, flexible.
6. Carry high information capacity.
7. Low loss
8. High data rate transmission.

n02-components

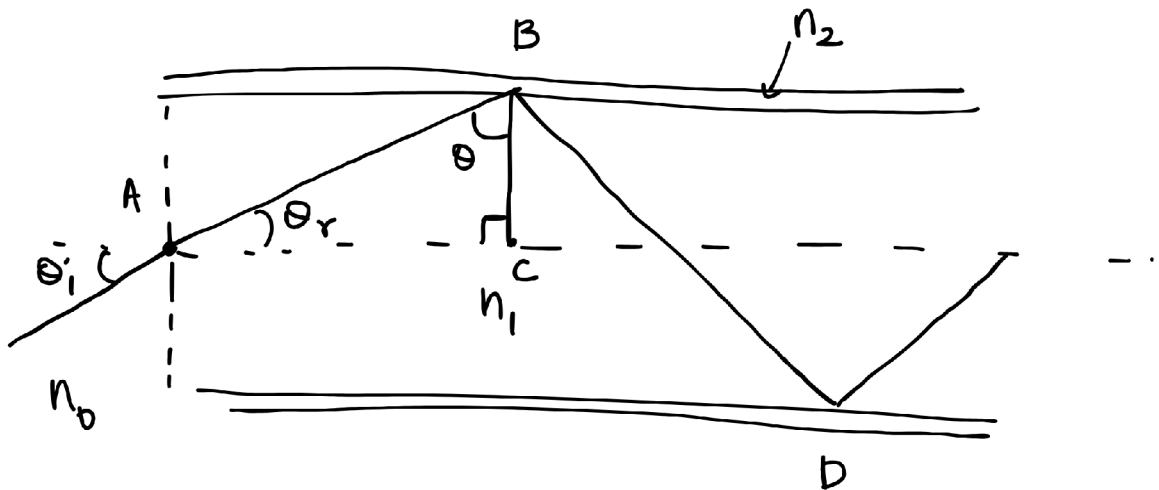


n03-guiding_of_light_through_optical_fibre

- Light can be guided through by TIR conditions for propagation of light through optical fibre.
- i) when incident light makes an angle greater than the critical angle at core cladding interface at the top of the fibre it will be reflected down to the core due to TIR.
- ii) when the reflected light when it makes an angle greater than or equal to θ_c . It will be reflected again up to due TIR at the bottom of optical fibre. Thus the propagation of light will be in the form of multiple total internal reflection through optical fibre.

(Diagrams showing $i < \theta_c$, $i = \theta_c$, $i > \theta_c$)

n04-numerical_aperture



- Consider a cross sectional view of optical fibre having core and cladding of refractive indices n_1 & n_2 . Where $n_1 > n_2$
- Let the fibre be in air medium whose refractive index is n_0
- The incident light while entering into core at A makes an angle of θ_i with the fibre axis.
- In core it travels along AB and is incident at point B at core-cladding interface
- Let θ_r be the angle of refraction at A and θ be the angle of incidence at B.
- When $\theta > \theta_c$ then TIR takes place in the and light takes the path BD.
- Due to multiple TIR propagation of light takes place through fibre.

Apply Snells law at A

$$n_0 \sin \theta_i = n_1 \sin \theta_r \dots (i)$$

From $\triangle ABC$,

$$\theta_r + \theta = 90^\circ$$

$$\theta_r = 90^\circ - \theta \dots (ii)$$

from (i) and (ii)

$$n_0 \sin \theta_i = n_1 \sin(90^\circ - \theta)$$

$$n_0 \sin \theta_i = n_1 \cos \theta \dots (iii)$$

At core-cladding, let θ_a be the maximum angle of incidence for which $\theta \geq \theta_c$ at core cladding interface to undergo TIR.

$$n_0 \sin \theta_a = n_1 \cos \theta_c$$

$$\sin \theta_a = \frac{n_1}{n_0} \cos \theta_c$$

$$\sin \theta_a = \frac{n_1}{n_0} \sqrt{1 - \sin^2 \theta_c}$$

$$\sin \theta_a = \frac{n_1}{n_0} \sqrt{1 - \frac{n_2^2}{n_1^2}} \quad (\text{since } \sin \theta_c = \frac{n_2}{n_1})$$

$$\sin \theta_a = \frac{n_1}{n_0} \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$\sin \theta_a = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

- Numerical Aperture is defined as the light gathering capacity of optical fibre.
- It is always proportional to the acceptance angle that is:

$$NA = \sin \theta_a = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 + n_2)(n_1 - n_2)}$$

where Δ is the fractional change of refractive index.

$$\Delta = \frac{n_1 - n_2}{n_1}$$

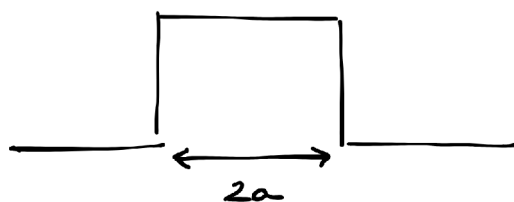
$$n_1 - n_2 = \Delta n_1$$

$$NA = \sqrt{(2n_1)(n_1 \Delta)} \quad (\text{Assuming } n_1 \approx n_2)$$

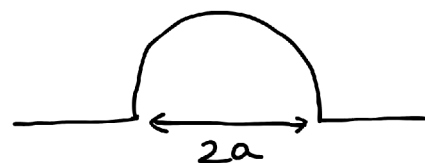
$$NA = n_1 \sqrt{2\Delta}$$

n05-types_of_optical_fibres

- Based on refractive index profile change, optical fibres are of two types



step-index fibre

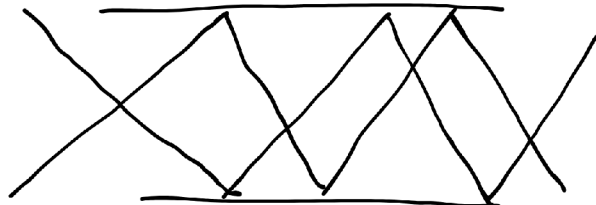


graded-index fibre

- Based on no. of paths available for a light to travel through fibre there are two types.



single-mode
fiber



multi-mode
fiber

n06-v_parameter

V parameter is defined as

$$V = \frac{2\pi}{\lambda} \cdot a \cdot \sqrt{n_1^2 - n_2^2}$$

$a \rightarrow$ radius of core

$\lambda \rightarrow$ wavelength

$n_1 \rightarrow$ R.I of core

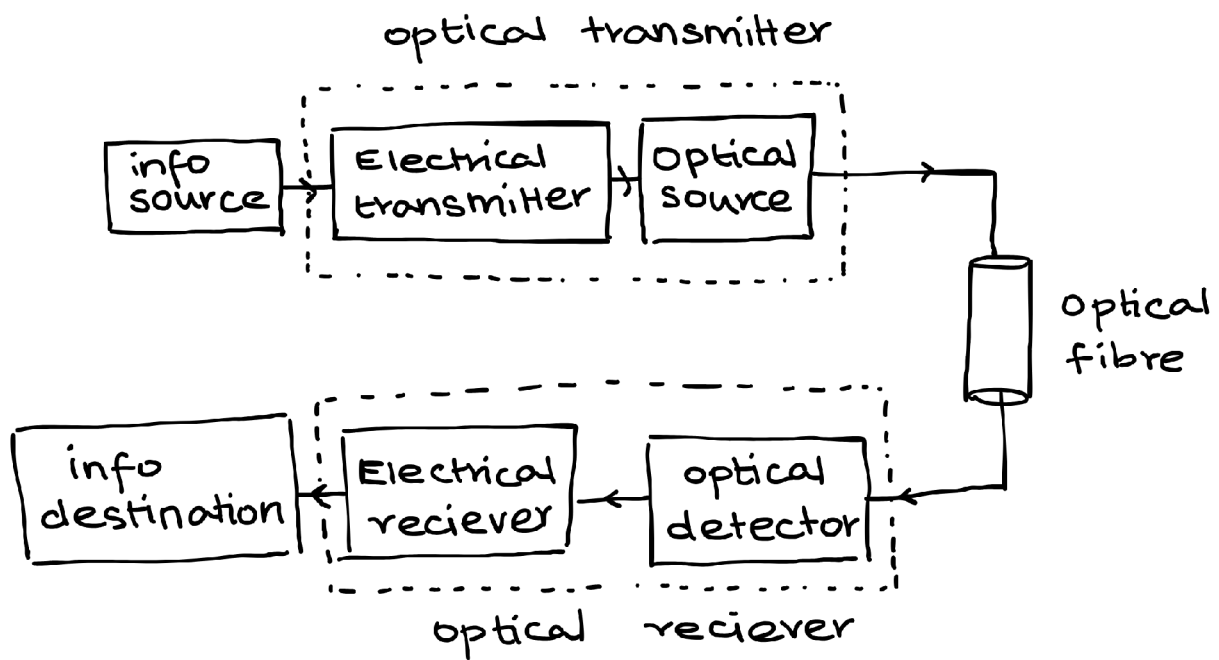
$n_2 \rightarrow$ R.I of cladding

- The no of modes in a step index fiber is $\frac{V^2}{2}$
- The no of modes in a graded index fiber is $\frac{V^2}{4}$

n07-fibre_optics_communication

It mainly consists of three parts

- Optical transmitter
- Information pathway
- Optical receiver



Ex: A multimode step index fibre with refractive indices of core and cladding 1.53 and 1.50. If the wavelength of light signal is $1\mu\text{m}$ and radius of core is $50\mu\text{m}$, find the no of possible modes through it.